

ROVARI

RV Circuit Studio

Quick Start Guide

rv-embedded.com

Getting Started

1. Plug in your CircuitPython board via USB.
2. The IDE detects the board and connects automatically.
3. Open code.py from the Project Explorer on the left.
4. Edit your code and click Run (green arrow) to save and run on the board.

Your code runs on the board and a backup copy is saved to your computer every time you hit Run.

Toolbar

The toolbar runs along the top, left to right:

Shortcut	Action
New File	Create a new empty file
Open Project	Open a project folder
New Tab	Open a new editor tab
Copy	Copy to clipboard
Save	Save current file
Save As	Save with a new name
Close Tab	Close current tab

Toolbar Letter Buttons

Shortcut	Action
E	Project Explorer (file browser)
S	Snippets panel
C	Camera view
T	Toggle bottom panel (REPL/Plotter)
D	Advanced Debugger side panel

Toolbar Action Icons

Shortcut	Action
Run (green arrow)	Save to board and run code.py
Stop (red X)	Send Ctrl+C to stop running code
Debug (bug icon)	Start the visual debugger
Settings (gear)	Open settings dialog
{ }	Snippets panel toggle
Libraries	Open library manager
Plotter	Switch to plotter tab

Keyboard Shortcuts

Shortcut	Action
Ctrl+S	Save current file
Ctrl+Shift+S	Save As
Ctrl+W	Close current tab
Ctrl+F	Find and Replace
Ctrl+.	Send Ctrl+C to board (stop program)
Ctrl+D	Send soft reboot (restart code.py)
Ctrl+/ Tab	Toggle comment on selected lines
Tab	Indent selected lines
Shift+Tab	Unindent selected lines
Ctrl+Scroll	Zoom editor font size

REPL

The REPL tab at the bottom shows serial output from the board. Type directly to send commands.

Shortcut	Action
Ctrl+C	Interrupt running program
Ctrl+D	Soft reboot (runs code.py)
Ctrl+E	Enter paste mode
Ctrl+V	Paste clipboard to the board
Up / Down	Previous and next command
Home / End	Jump to start or end of the line
Right click	Copy, Paste to Board, Clear

Serial Plotter

The Plotter tab graphs numeric serial output in real time.

Supported Print Formats

```
print((val,))    Single value tuple
print((x, y, z)) Multi value tuple
print("a={},b={}".format(a,b))  Labelled values
print(a, b, c)   Space separated
print(f"{a},{b},{c}") CSV
```

Plotter Controls

Shortcut	Action
Window (spinbox)	Number of samples visible (default 500)
Pause / Resume	Freeze or resume live plotting
Record	Start capturing data to buffer
Stop Rec	Stop recording, auto-saves CSV
Analyze	Open recorded data in analysis view
Open CSV	Load a CSV file into analysis view
XY	Toggle parametric mode (first column = X axis)
Clear	Remove all traces and reset

Recorded data auto-saves to ~/CircuitStudioData/ as a timestamped CSV.

Analysis View

Opened from Analyze or Open CSV. Full interactive plot with:

- Scroll wheel to zoom
- Click and drag to pan
- Ccrosshair shows coordinates and nearest data point
- Click trace checkboxes to show/hide channels.
- Stats bar shows min, max, avg, std, and sample count per channel.
- XY checkbox for parametric view
- Auto Fit resets zoom.
- Export CSV to save.

Visual Debugger

1. Click the Debug icon (the little bug) on the toolbar.
2. The Advanced Debugger panel opens on the right by clicking D, select files to instrument.
3. Click in the gutter beside any line to set a breakpoint. A red dot appears.
4. Add watch expressions in the side panel, one per line, e.g. counter, sine, buf[0:4]
5. Add conditional breakpoints if you want them, e.g. counter > 10
6. Click the Debug icon again to start. The board halts on the first line.
7. Click Run to continue to your next breakpoint, or Step to advance one line.

Breakpoints

Click the gutter, the narrow strip left of the line numbers, to set or clear a breakpoint. Breakpoints are stored by the IDE, not written into your file, and they are remembered when you close a tab or restart the IDE.

Breakpoints go on executable lines. Clause lines such as elif, else, except and finally will not accept one, because there is nowhere to place the debug code. Put the breakpoint on the first line of the body instead. It fires whenever that branch is taken, which tells you the same thing.

Debug Controls

Shortcut	Action
Run (green)	Continue execution
Step (blue)	Execute one line and stop
Restart (grey)	Restart from the top
Stop (red)	End debug session

The executing line is highlighted in the editor with a blue bar, watch values update in the debug bar and side panel.

Library Manager

Click Libraries in the toolbar. You have two options:

1. Auto Install: the IDE scans your code.py imports, finds matching libraries in the Adafruit Bundle, and installs them to CIRCUITPY/lib/ with one click.
2. Browse: search and install individual libraries from the full Adafruit and Community bundles.

Snippets

Click S to open the snippets panel. Double click any snippet to open it as a complete sample in a new tab. Categories include GPIO, debounce, state machines, ADC/DAC, PWM, DSP filters, serial protocols, and more.

Settings

Click the gear icon. Available settings:

Shortcut	Action
Editor Font Size	Code editor zoom level
UI Font Size	REPL, file explorer, tabs, snippets
Tab Width	Spaces per tab (default 4)
Vim Mode	Enable Vim keybindings
Autocomplete	Toggle code completion
Line Numbers	Show or hide line numbers
Board Filename	File to save (default code.py)
Auto Connect	Connect to board on detect
Auto Save	Save before running

Settings are saved per user, not inside the application, so they persist across updates:

Windows: %LOCALAPPDATA%\RV Circuit Studio\

macOS: ~/Library/Application Support/RV Circuit Studio/

Linux: ~/.local/share/RV Circuit Studio/

Troubleshooting

My Board is not detected

Check the USB cable is a data cable, not power only. Try a different USB port and make sure CircuitPython is installed on the board (circuitpython.org).

I Cannot save to board

The board filesystem may be read-only. Check boot.py for storage.remount() calls. Double tap the reset button to enter safe mode, then delete or fix boot.py.

The Plotter shows no data

Make sure the plotter is resumed (not paused). Your code must print numeric data, check the REPL tab to confirm output is printing.

The Libraries are not installing

Make sure the board is connected and the CIRCUITPY drive is visible. Check available space on the board. Some libraries require .mpy files which the manager installs automatically.